



CREDIFY: contextualized retrieval of evidence for open-domain fact verification

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Abstract

Verifying the credibility of claims made in public speeches and articles is crucial in the current digital era due to the widespread prevalence of fake news and its impact on society. Finding authentic evidence to verify these claims efficiently is necessary to counteract their harmful dissemination. To meet this objective, journalists manually verify the claims, which is challenging. Automated claim verification frameworks have emerged as a potential solution to this problem, gaining significant attention in recent years. These frameworks provide an efficient solution to ensure the credibility of information. However, a significant challenge lies in retrieving the most relevant, consistent, and authentic evidence against these claims that considers the context and predicts their veracity. To this end, we propose a contextualized retrieval of evidence for the open-domain fact verification (CREDIFY) framework. This automated open-domain claim verification architecture utilizes context and semantics to retrieve consistent evidence written by journalists and experts. This verified evidence serves as a reliable source for predicting the veracity of claims. CREDIFY consists of three modules: *passage retrieval*, *sentence selection*, and *veracity prediction* for retrieving contextual evidence and performing entailment classification for veracity prediction. To validate our methodology, we propose the retrieval and entailment of claims through inference for the facts' testimony dataset collected from fact-checking sites. It is composed of 5000 open-domain claims supporting or refuting 3356 consistent and authentic evidence verified by journalists. The proposed framework achieves a significant performance gain on end-to-end systems with a margin of 5.5% over the state-of-the-art work, achieving an accuracy of 96.5%. Moreover, we also analyze the effect of the contrastive claim generation and paraphrasing technique on the veracity prediction module for de-biasing and generalizing our fact verification model. The code for our proposed model, CREDIFY, is available on GitHub <https://github.com/SarahNasir54/Credify.git>.

Keywords Fact · Claim verification · Evidence · Evidence retrieval · Transformers · Sentence selection · Veracity prediction

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1 Introduction

In recent years, the exponential growth of digital content has led to a corresponding increase in the spread of misinformation, fake news, and manipulated facts [1]. Popular platforms *the Onion*, *World News Daily Report*, *Nation Report*, and political groups repeatedly make false claims [2]. These claims are also made by famous public figures in their speeches and interviews [3]. Although fake news has been a long-standing problem since the establishment of the printing press, the problem has become more serious with various digital media outlets and news blogs. These online media platforms generate vast information with possibly inaccurate and misleading news. Anyone with access to these platforms can write fake articles and spread rumors about politics, health, finance, etc. [4]. The motivation behind fake news is to damage the reputation of a renowned person, product, or organization and spread violence in society [5].

Although fake news is not a new topic, its negative influence on the outcome of the US elections in 2016 brought it to the forefront of public attention [6–8], and the number of fact-checking organizations increased from 44 to 114 in just two years.¹ The ultimate objective of spreading fake news during elections was to change voters' perspectives against their opponents [9]. After facing the disasters in these elections, the COVID-19 pandemic has undoubtedly highlighted this problem again, where misinformation poses a significant challenge to public health and safety [10]. Fact-checking organizations play a significant role in the battle against the spread of false information. Trained and professional journalists who are already aware of the context of fake news or rumor [11], put a fair amount of time into examining the truth by finding authentic sources, conducting deep investigations, and then propagating the veracity through various platforms, including instant messaging services, online social networks, and the media. Popular Fact-Checking organizations such as PolitiFact,² FactCheck,³ FullFact.org⁴ and Snopes⁵ perform this whole verification manually [12], which is challenging because manual detection of facts requires a lot of resources, effort and time [13]. These challenges motivated researchers to develop automated fact verification models that autonomously retrieve evidence of claims and predict their veracity [14].

Significant progress can be seen in developing automated fact-verification models and benchmark datasets [15, 16]. These frameworks perform the task of verifying and predicting the veracity of claims in three modules: *passage retrieval*, *sentence selection* and *veracity prediction* [17–20]. In the passage retrieval module, the model retrieves the evidence passage against a claim query from top web search results or locally stored corpus containing experts' and journalists' verified facts. The next module, sentence selection, extracts the most relevant sentence from the retrieved passage that strongly helps predict the claim's veracity. This relevant sentence acts as final evidence for verifying the claim. The final module—claim veracity prediction—is the core component of the automated fact verification model. This module uses natural language inference (NLI) techniques to determine if the extracted evidence sentence supports, refutes, or does not provide the information regarding the query claim. Figure 1 shows the process of verifying claims with an automated fact verification system.

¹ <https://reporterslab.org/fact-checking/>.

² <https://www.politifact.com/>.

³ <https://www.factcheck.org>.

⁴ <https://fullfact.org>.

⁵ <https://www.snopes.com/>.

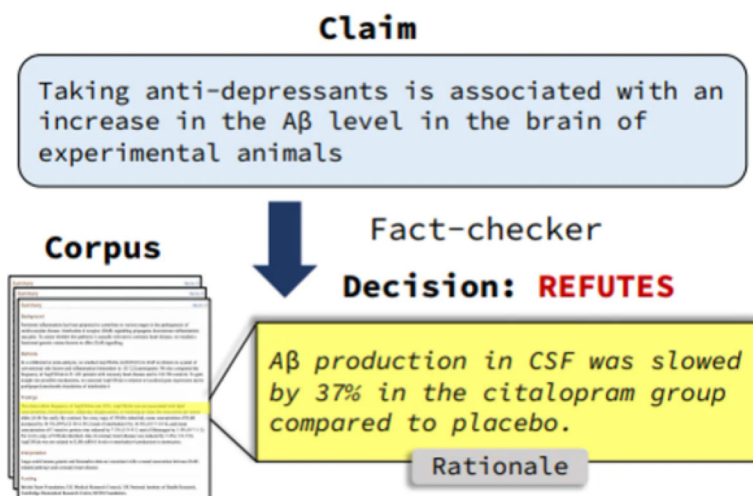


Fig. 1 A general fact verification system retrieving the evidence from a corpus and predicting the claim veracity decision as refute

The previously proposed datasets focused on collecting evidence from inconsistent top web search results [21, 22], unreliable knowledge sources like Wikipedia [17, 23, 24], single biased platforms [21, 25], or belongs to only one domain, for example, health [26], science [27] and finance [28] which are not suitable for verifying diverse topic claims. Also, the previously used evidence retrieval techniques in automated fact verification architectures either use BM-25 [29], TF-IDF [15, 30], keyword matching, which does not consider context and semantics-based retrieval or dense passage retrieval (DPR) [30] models which heavily rely on large-scale training data to learn representations of passages and claim query [31, 32]. The limitations present in current automated fact-checking datasets and contextual-based evidence retrieval techniques for verifying the claims still need attention.

In this research, we emphasize the problem of verifying topic-enriched open-domain claims with consistent evidence, leading to the development of a context-aware transformer-based architecture. We present CREDIFY architecture that, on the one hand, assesses the veracity of the claims with contextual evidence against a corpus of expert-written authentic information. On the other hand, we provide a novel dataset, RECTIFY, composed of 5000 open-domain claims supporting or refuting 3356 consistent pieces of evidence verified by journalists and professionals. Our proposed CREDIFY architecture has three modules: passage retrieval, sentence selection, and veracity prediction. In the first stage, our architecture CREDIFY extracts the passage from the evidence corpus through a transformer-based model designed explicitly for semantic-based passage retrieval. In the second stage, the top relevant sentence is extracted. The following module then uses it to predict the veracity of claims using pre-trained NLI techniques. CREDIFY uses an evidence retrieval technique grounded in contextual-based methodologies, leveraging a powerful transformer model to enhance the effectiveness of our approach. We also analyze the effect of data augmentation (contrastive claim generation) at the inferencing module by de-biasing our dataset and analyzing the impacts on our proposed architecture.

In contrast to previous studies, RECTIFY relies on collecting open-domain claims from reputable fact-checking sites with authentic evidence stored in a corpus verified by domain experts. This technique of maintaining a real-world dataset ensures the high-quality predic-

tions generated by the model. Another major conflict lies in utilizing the powerful transformer mechanism, which is pivotal in retrieving pertinent contextual evidence and improving the overall claim veracity prediction process. Moreover, our architecture maintains a consistent corpus of evidence to avoid factual inconsistency problems instead of relying on web search results, thus discouraging the real-time retrieval of evidence.

With CREDIFY, we aim to facilitate manual fact-checkers by developing an architecture that retrieves contextual evidence for verifying multi-domain claims from a consistent evidence corpus rather than relying on top web search results. Our proposed architecture outperforms state-of-the-art models, providing fact-checkers with a more effective model for an accurate and efficient verification process. Moreover, CREDIFY significantly improves existing techniques [15, 30] with pre-trained transformer models on the module level. To our knowledge, no previous research has proposed an end-to-end automated architecture with all these characteristics with transformer-based models. CREDIFY attains significant performance gain with a margin of 5% over the state-of-the-art work, achieving an accuracy of 96.5%.

The contributions of the paper can be summarized as follows:

- We propose a new dataset, RECTIFY, which consists of open-domain claims with their justification articles as evidence for accurate prediction of veracity labels.
- We propose a complete end-to-end architecture, CREDIFY, which relies on automated context-based evidence retrieval techniques and predicts claim veracity.
- Integration of a data augmentation technique with contrastive claim generation and paraphrasing for de-biasing and generalizing our proposed model.
- The proposed methodology attains a performance gain on end-to-end systems with a margin of 5.5% over the state-of-the-art work, achieving an accuracy of 96.5%.

The remaining paper is structured as follows: Sect. 2 covers the relevant literature for the automated fact verification model. Section 3 presents the proposed dataset in detail. In Sect. 4, the proposed methodology is discussed. Experiments are presented in Sect. 5. The results of the experiments are reported in Sect. 6. Finally, Sect. 7 concludes the research study with future directions.

2 Related work

In this section, we review the previous studies conducted on automated fact verification. We discuss the end-to-end architectures that retrieve evidence from web [22] or the local repository. The techniques for retrieving evidence can either take into account term frequency [17, 27], contexts [33], semantics [26], or relevancy score [19]. Finally, we present the prominent studies on entailment classification, which predicts the veracity of the claim using neural network architectures [33], stance detection [29], and pre-trained NLI models [18, 20, 34]. Table 1 overviews the studies most relevant to our research work.

To mitigate the effects of fake news, Hassan et al. [2] proposed the ClaimBuster model, the first automated end-to-end system for verifying claims. They collected checkworthy claims from politicians' speeches and interviews. The extracted claims were then verified with the facts in an internal knowledge base written by professional journalists. The techniques used for matching the claim were parts-of-speech tags and entities. The claim queries were sent to the web for retrieving evidence if it was not found in a local repository. Their study lacks the utilization of advanced deep-learning neural network models and NLI techniques.

Table 1 Comparison table of state-of-the-art fact verification models

Ref	Model	Evidence retrieval	Textual entailment	Results	Dataset
Thorne et al. [17]	MLP based fact verification	TF-IDF	MLP	Accuracy 31.87%	FEVER
Liu et al. [18]	VERISCI	TF-IDF	RoBERTa- Large	Precision 47.5%.	SciFact
Kotonya and Toni [26]	Health Fact check	S-Bert	RoBERTa- snli	Accuracy 69.73	PUBHEALTH
Bouziane et al. [19]	FABULOUS	BM25	MLP	Accuracy 63.64 %	FEVEROUS
Samarinas et al. [34]	Quin+	BERT	RoBERTa- Large	F1 score 58.22%	FACTUAL- NLI+
Gupta et al. [20]	Dialogues fact checking	TF-IDF	Aug-wow	Accuracy 63.9%	DIALFACT
Khan et al. [30]	Articles based fact checking	DPR	RoBERTa-base	Macro F1 score 82.1%	WatClaim Check
Rangapur et al. [28]	Financial claims fact checking	–	GPT-2 model	Accuracy 47%.	Fin-fact
DeHaven [15]	BEVERS	TF-IDF and NER	RoBERTa- Large MNLI	Accuracy 80.24%	FEVER

This lack of claimbuster dataset was covered by Thorne et al. [17]. The researchers in this study proposed the FEVER dataset consisting of 185,445 claims and evidence. These pieces of evidence were collected from an external knowledge source, Wikipedia, and then stored locally in a corpus. Their proposed pipeline was divided into three stages: relevant document retrieval, relevant sentence selection, and textual entailment. The TF-IDF technique was used for retrieving relevant documents and sentences. Finally, textual entailment was performed using a multi-layer perceptron (MLP). The claims were classified into support, refute, and insufficient information. Their model achieved an accuracy of 31.87%. Similarly, Reddy et al. [35] DeFactoNLP model, a four-stage pipeline: document retrieval, sentence selection, entailment classification, and score classification. Named entity recognition was used for retrieving documents, and top sentences were selected based on keyword matching and TF-IDF. The pipeline was implemented on the FEVER-SNLI dataset and achieved an F1 score of 51.36%. The approaches used in the above studies lack semantics and contexts in retrieval stages, so the results affect the performance of veracity prediction.

Stance detection is another approach for predicting the claim's veracity. Nadeem et al. [29] used stance detection for predicting claim veracity on the FEVER dataset. The model predicted the factuality with the assistance of evidence provided at the sentence and document levels. Their end-to-end system worked with four steps: document retrieval, stance detection, evidence extraction, and linguistic analysis. The stance detection was applied on high-ranked documents, and the rationales were calculated at the sentence level. Finally, they performed linguistic analysis. Their model gave an accuracy of 59.35%. Similarly, using BiLSTM and XLM model with transfer learning achieved an F1 score of 28% for a cross-lingual fake news detection task [36].

The studies discussed above rely on Wikipedia to verify general claims. However, it is not worth considering a reliable platform [17] for evidence collection due to the possibility of manipulated information. To verify domain-specific claims with authentic expert evidence, Wadden et al. [27] proposed the sciFact dataset, which consisted of 1,409 expert-written articles and claims. Their three-step methodology, abstract retrieval, sentence selection, and veracity prediction, verified scientific claims from published scientific articles. They trained their model with FEVER, UKP, and Snopes datasets, followed by sciFact fine-tuning. Their model gave a precision of 47.5% with RoBERTa-large. Similarly, Liu et al. [18] extended the previous work by implementing their methodology on the sciFact dataset and proposed a SciKGAT model. They used TF-IDF to extract the most relevant passage and sentence. Further, they used the KGAT model and then predicted the veracity with the RoBERTa model. Their model gave a precision of 84.26%. Likewise, Pradeep et al. [37] used the T5 language model and implemented their pipeline on the sciFact dataset. They used monoT5 ranker and T5-3B for sentence selection and veracity prediction. Their model achieved an F1 score of 77.47%.

After the emergence of verifying domain-specific claims, Kotonya and Toni [26] proposed another automated fact-checking model that verifies public health claims specifically. They constructed a new dataset, PUBHEALTH, which consists of 11.8k claims and justifications provided by experts and journalists to support claims. They performed two tasks on their dataset: veracity prediction and explanation generation. They first used the SBERT model for sentence-pair ranking and then applied softmax for veracity prediction. Moreover, they used a BERT-based summarization model pre-trained on the Dailymail and CNN news articles and summaries for explanation generation. The results showed that SciBERT for veracity prediction outperforms with an accuracy score of 69.73%, and the EXPLAINER FC-EXPERT model for explanation generation gave the highest ROUGE-1 of 32.30. Truică et al. [38] proposed a hybrid approach called DANES which effectively combined textual and social context

embeddings for fake news detection. DANES utilized Mittens for larger datasets and FastText for smaller datasets to capture richer textual representation. They used social network features by encoding user interactions, enhancing performance with an accuracy of 79%. Recent studies have explored the impact of document representation on clustering and topic modeling. Embeddings such as Doc2Vec proved to be better than traditional TFIDF for clustering [39]. Doc2Vec effectively captured semantic relationships while reducing dimensionality to enhance the performance of clustering algorithms. Similarly, density-based methods such as DBSCAN and centroid-based algorithms such as K-means and Spherical K-means showed significant improvement in document clustering. DenLAC another clustering algorithm used density-based clustering to handle diverse clusters [40].

Bouziane et al. [19] proposed a novel approach to fact-checking, leveraging language understanding techniques to analyze unstructured and structured information. They proposed two methods for handling multi-hop multi-model datasets: robust retrieval of evidence sets and noisy evidence input handling. BM25 technique was used for efficient document retrieval, measuring a document's relevance score against the query. The multi-hop density retrieval (MDR) technique was used for passage retrieval. They proposed noisy entailment through the adapted training (NEAT) method. The above pipeline was implemented on a FEVEROUS dataset, which gained 63.64% accuracy using the MLP model. However, the results were ineffective as BM25 does not represent contextual representations well. Apostol et al. [41] proposed a distributed system for real-time hazard event monitoring by detecting geolocation content communities on X using topic modeling to identify misinformation. To handle data imbalance issues in topic modeling, Truica and Leordeanu [42] introduced a classification solution containing information about defects of different tree approaches. They used decision trees, random forest, and extremely randomized trees for classification, whereas Truica et al. [43] used an automatic term recognition (ATR) to extract domain-specific terms which improved topic coherence and readability.

For evidence retrieval, Sathe and Park [33] proposed a model by considering the whole document as evidence with BERT and multiple instance learning. They used a pre-trained SBERT architecture for evidence retrieval. In support verification, the claim and the retrieved sentences used BERT and multiple instances learning, and finally, softmax was applied to predict the claim's veracity. The results showed that SBERT+BERT achieved an accuracy and F1 score of 81.9% and 81.0%. Samarinas et al. [34] also proposed a step-down approach, namely Quin+, which focused on verifying open-domain claims. The model consisted of three-stage pipelines: passage retrieval, sentence selection, and entailment classification. They implemented their pipeline on their proposed factual NLI+ dataset incorporating synthetic examples from question-answering datasets and noise passages from web search results. The passage retrieval worked with a QR-BERT model, which first calculates token embedding and then calculates relevance between the claim and evidence using the dot product. Sequence labeling and embedding-based techniques were used for sentence selection. They trained RoBERTa-large with a linear transformation of the [CLS] token embedding for entailment classification. Their approach gave an F1 score of 58.22% with a sequence labeling sentence selection technique.

Likewise, Sadeghi et al. [44] focused on verifying open-domain claims by extending the LIAR and FakeNewsNet datasets. They used machine learning and deep learning word embedding techniques and achieved an accuracy of 85.58% and 41.31%. Their study lacks the utilization of advanced transformer models. Constructing a large-scale corpus is another challenge in automated fact verification. To address this challenge, Pan et al. [45] proposed a QACG model that assisted them in generating claims. The model generated claims that can be supported or refuted with the help of the information available on Wikipedia. They first

generated question-answer pairs from the evidence and converted them into claims. They used the FEVER dataset and RoBERTa zero-shot model for fact verification and achieved an F1 score of 77.3%. However, the model's reliance on automatically generated claims could lead to the propagation of errors or biases in the training data due to dependence on one source.

Despite the criticisms and acknowledged weaknesses associated with Wikipedia as a reliable source, it remains a focal point of recent research. For example, Gupta et al. [20] constructed a DialFact dataset for annotated conversational claims, which were paired with evidence from Wikipedia. Three sub-tasks were performed in DIALFACT: claim detection, evidence retrieval, and claim verification. Their model gave an accuracy of 63.9%. Existing fact-checking models trained on non-dialogue data like FEVER fail to perform well on their task. DeHaven [15] proposed the BEVERS model, a three-step fact verification pipeline. The pipeline was implemented on FEVER, SciFact, and PUBHEALTH datasets. The objective of their work was to tune hyperparameters for optimal performance. The evidence for all three datasets was stored in a local repository instead of retrieved from web results. They used a simple TF-IDF and NER (Named Entity Recognition) for evidence retrieval. The Roberta-Large MNLI and DeBERTav3-mnli-snli-anli were used for entailment classification. Their model gave a label accuracy of 80.24% on the FEVER dataset. Since their retrieval model is based on term matching, semantic representations and word embedding techniques could perform better.

On the contrary, current research also focuses on constructing automated fact-checking, which uses trusted information sources like fact-checking sites to retrieve evidence and make an accurate decision regarding the claim's veracity. For this, Khan et al. [30] proposed a two-step automated fact-checking pipeline: sentence selection and entailment classification. The pipeline was implemented on their proposed dataset WatClaimCheck, which contains the claims made in interviews, speeches, and social media. The premise articles' URLs were used as an evidence source. The techniques used for sentence selection from premise articles are TF-IDF and DPR. Also, the RoBERTa-Base model was used for entailment classification. The loophole in their research is that the verdict labels against the claim were categorized into 86 classes, which they later converted into three without defining any mapping mechanism. So, their dataset does not reflect the real-world tasks of training models. Their study shows that their framework benefits from significant training data to achieve optimal performance. The growing reliance on large language models is also seen as a developing trend in automated fact verification. Rangapur et al. [28] used GPT-2 for textual entailment for financial claim veracity prediction. Also, the GPT-3 model was used for multidomain scientific claim verification [16]. Their model achieved an F1 score of 61% when provided with zero supervision.

Word embeddings play a significant role in NLP, context-aware word embeddings such as Word2Vec, FastText, and Glove proved to be beneficial in veracity tasks. Ilie et al. [46] employed recurrent convolutional neural network-based architecture using Word2Vec, FastText and Glove embeddings, where Word2Vec with skip-gram model achieved a 20% higher accuracy than the other two word embeddings. These static word embeddings were not beneficial as they did not capture the contextual information in the text. Transformer-based embeddings such as BART and RoBERTa proved to perform better in the misinformation detection task [47]. The BiLSTM-CNN model with BART embedding achieved a higher accuracy of 92.50%. Transformer-based embeddings also performed well with simple neural networks [48]. It proves that embeddings play an important role in any NLP task and used BiLSTM and BiGRU using document embedding with the BART model which outperformed other models.

Network immunization is a crucial task in network analysis, which helps to prevent the spread of harmful content on social networks. Petrescu et al. [49] proposed an immunization method called SparseShield to scale existing immunization while minimizing harm to user experience by demoting harmful content. Later on, a tree-based mitigation strategy, MCWDST [50], was introduced that leverages deep learning for fake news detection. This approach immunized nodes in real-time through a minimum cost-weighted directed spanning tree. Similarly, Apostol et al. [51] introduced CONTAIN, a community-based algorithm that detects harmful content spreaders and ranks network partitions for immunization. CONTAIN outperformed Netshield and SparseShield in both speed and efficiency. While network immunization is useful in misinformation detection and harmful content dissemination in online social networks, our method focuses on claim verification on news websites which does not include networks. Rădulescu et al. [52] improved Louvain's community detection algorithm by connecting the nodes using WordNet lemmatizer and document embeddings.

The literature review reveals a notable gap in automated fact verification systems. Specifically, there is a need for a system capable of verifying claims across different domains while maintaining a consistent corpus of evidence to avoid factual inconsistency problems rather than solely relying on popular web search results. Additionally, existing evidence retrieval techniques often fall short. They typically rely on simplistic methods like keyword matching or relevance scores, which overlook the nuances of semantics. Even advanced models like DPR face limitations in achieving optimal performance due to their dependence on extensive training data. In essence, what is lacking is an evidence retrieval technique that harnesses the power of transformer-based methods. Such a technique would consider the semantics and context of claims requiring less training data, thus enhancing the credibility of information verification processes.

To overcome these limitations, our research introduces a novel solution for addressing these challenges: the CREDIFY model and the RECTIFY dataset. These innovative techniques have been purposefully developed to overcome the identified limitations. The CREDIFY model employs advanced techniques for enhancing evidence retrieval by considering semantics and context for veracity prediction. At the same time, the RECTIFY dataset offers diverse open-domain claims with a consistent evidence corpus. Together, these advancements aim to boost the efficiency and reliability of automated fact verification models, thus significantly contributing to the research field. Section 3 describes the RECTIFY dataset construction process, and Sect. 4 discusses the proposed methodology in detail.

3 RECTIFY dataset

This section presents an overview of the RECTIFY data curation process, including data collection and exploratory data analysis (EDA). First, we highlight the process for data collection, including sources and data extraction procedures. Following this, we provide a detailed analysis of the dataset statistics, including the details of the training, testing, and evidence passages.

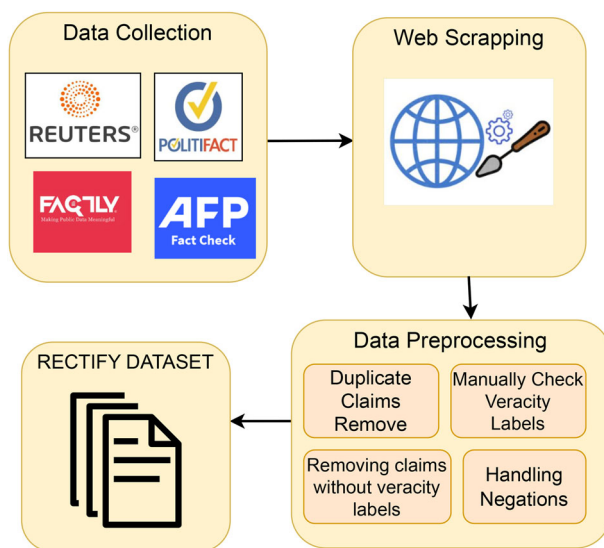


Fig. 2 RECTIFY data curation process

3.1 Data collection

We collect claims and news from multiple domains with their justification articles collected from popular fact-checking sites: Snopes,⁶ PolitiFact,⁷ AFP,⁸ Reuters,⁹ FactCheck.org,¹⁰ and Google fact check.¹¹ These claims comprise a diverse range of subjects and topics. Moreover, we also construct a consistent evidence corpus for verifying and predicting the veracity of the claims that provides all the necessary contextual evidence. We use Google fact check API and web scrapper tool¹² to collect metadata from fact-checking websites. We manually extracted the justification articles that these fact checkers provided as evidence supporting the claim. Most of the articles usually end up with sections with a heading *our ruling*, *summing up*, *if your time is short* and *conclusion*. These sections usually have sentences that are strongly related to a claim and help in predicting the veracity of the claim. We extract the first five sentences from the summary and conclusion section from the claim justification article as this is the approach used by the LIAR-PLUS dataset [25]. We filter out and merge these sentences as a passage and maintain all the evidence in a corpus. This evidence passage is retrieved to automatically predict the veracity label by analyzing whether the evidence supports or contradicts the claim. We manually removed duplicate instances and the claims without veracity labels. We handle negations by removing unnecessary punctuation and assigning correct veracity labels to claims. Figure 2 describes the data generation steps in detail. The collected dataset serves as the backbone of our fact verification task, which helps

⁶ <https://www.snopes.com/>.

⁷ <https://www.politifact.com/>.

⁸ <https://factcheck.afp.com/>.

⁹ <https://www.reuters.com/fact-check/>.

¹⁰ <https://www.factcheck.org/>.

¹¹ <https://toolbox.google.com/factcheck/explorer>.

¹² <https://webscraper.io/>.

Table 2 Top 15 most common news and fact-checking websites with the count of articles from each source

Popular websites	Count
www.politifact.com	806
www.factcheck.afp.com	672
www.checkyourfact.com	464
www.factly.in	350
www.sochfactcheck.com	300
www.indianexpress.com	290
www.foxnews.com	107
www.geo.tv	102
www.indianexpress.com	65
www.boomlive.in	62
www.foxnews.com	50
www.thehindu.com	39
www.economictimes.indiatimes.com	36
www.aljazeera.com	34
www.dawn.com	31

Table 3 Example instance from our proposed dataset

Claim: "Putin bans 5 G across Russia, destroys all towers."

Justification Article: Fabricated story about Vladimir Putin banning 5 G in Russia repackaged on social media. Real Raw News' website includes an "About Us" page with a disclaimer that says it contains "humor, parody, and satire," and PolitiFact has debunked many of the fantastical, false stories the site has published. We rate this post False as We found no evidence that Putin has banned 5 G technology or destroyed 5 G towers in Russia. In 2019, Russian network RT America aired a segment warning about 5 G's supposed dire health threats, The New York Times reported, even as Putin "ordered the launch of Russian 5 G networks in a tone evoking optimism rather than doom."

Verdict: False

Date: August 15, 2023

URL: <https://www.reuters.com/article/factcheck-5g-russia-idUSL1N39Q271/>

us automate the veracity prediction process of claims. The dataset covers multiple topics gathered from popular sources, as illustrated in Table 2.

3.2 Dataset statistics

Our dataset consists of 5000 claims that can be verified against an evidence corpus of 3356 passages extracted from justification articles written by expert and professional journalists. We also provide the metadata for each claim in the dataset, which includes *veracity label*, *reviewer site*, *date*, and *URL*. Our dataset also includes gold standard evidence, which helps in assessing the performance of the evidence retrieval model. The veracity label in our dataset consists of three labels and refers to the original rating assigned by the fact-checking organization. In this labeling scheme, claims are assigned to one of the following categories, and their label distribution can be visualized in Fig. 3a:

4. The highlighted challenge underscores that utilizing fact-checking datasets trained on one specific domain (e.g., health or politics) may yield inaccurate or less effective results when applied to a different domain [26, 27].

To address these issues, our corpus overcomes the challenge of low-quality annotations by maintaining labels of the same groups assigned by the original fact-checking organization. Also, despite relying on one fact-checking site like Snopes or PolitiFact, we collect data from a diverse range of authentic sources that promote a more balanced dataset by reducing bias. Moreover, we have constructed a consistent corpus of evidence, which reduces the risks associated with retrieving inconsistent evidence from web sources, enhancing the integration of results. Furthermore, as described above, we have collected data from multiple domains. Hence, this diversification of data mitigates the domain mismatch problem, enabling the model to make more generalized, accurate, and practical fact-checking assessments. The comparison of the proposed dataset with existing fact-checking datasets is described in Table 5.

4 The proposed CREDIFY framework

Our proposed framework, CREDIFY, intends to help during the automated verification process by analyzing the claims circulating in online articles, blogs, and news platforms. CREDIFY implements an interconnected, end-to-end architecture with powerful transformer models trained on multidomain claims. CREDIFY reflects the real-world task of verifying claims collected from popular fact-checking sites with experts' written evidence stored in a consistent corpus, significantly different from previous automated fact-checking approaches. Moreover, another significant milestone has been achieved by implementing contrasting augmentation techniques for generalizing and de-biasing fact verification datasets. Our framework implements an interconnected, end-to-end architecture consisting of three modules: *passage retrieval*, *sentence selection*, and *veracity prediction* for performing evidence-based automated fact verification as shown in Fig. 4.

The first module retrieves the most relevant evidence passage from a large corpus of justification articles written by domain experts, journalists, and professionals. The second module extracts the most relevant sentence to claim from the retrieved passage, which acts as final evidence for predicting veracity. Finally, the last module uses entailment classification for predicting the veracity label by analyzing the relationship between the query claim and retrieved evidence sentence from the corpus. The entailment classification approach classifies the veracity label into one of three categories: support, refute, and NEI. The support label signifies that the evidence corroborates the claim, while the refute label indicates that the evidence contradicts the claim. Lastly, the label NEI denotes that the evidence fails to supply sufficient information to determine the claim. The overall flow of CREDIFY and the working of individual modules are described in detail in the below subsections.

4.1 CREDIFY modules

This section describes the details and techniques used in three modules of CREDIFY for performing automated claim verification.

Table 5 Proposed dataset compared with other existing fact-checking datasets

Dataset name	Claim source	Data size	Labels	Domain	Evidence source
Snopes	Snopes website	4956 claims	2	Open	30 google documents
MultiFC	Fact checking websites	34,918 claims	40	Open	Web search results
Claim buster	Speeches, debates, interviews	23,533 claims	3	Politics	Web search results
FEVER	Wikipedia	185,445 claims	3	Open	50,000 Wikipedia documents
Climate- FEVER	Wikipedia	1535 claims	4	Climate	7675 Wikipedia documents
Wiki fact check	Wikipedia	124,821 claims	2	Open	277,194 Wikipedia documents
PolitiFact	PolitiFact website	106 claims	3	Politics	Wikipedia articles
Perspectrum-Master	Debates	907 claims 11,164 perspectives	3	Human Perspectives	8092 passages from debate websites
SCI-Fact	SZORC	1409 claims	3	Scientific papers	5183 abstracts from research articles
Factual- NLI+	Fact checking sites	N/A	3	Open	Web search results
PUB- HEALTH	Fact checking sites	11,832 claims	3	Health	500+ documents from journal articles
Factify	Indian and U.S websites	50,000 claims	5	Politics and Covid-19	web search results
LIAR- PLUS	PolitiFact	12,836 claims	6	Politics	Wikipedia documents
Feverous	Wikipedia pages	87,026 claims	3	Open	50,000 Wikipedia documents
WatClaim check	Fact checking sites	33,697 claims	86	Open	Justification articles from multiple websites
HealthFC	Web portal Medizim transparent	750 claims	3	Health	Clinical studies
Proposed (RECTIFY)	Fact checking sites	5000 claims	3	Open	3356 expert written justification article

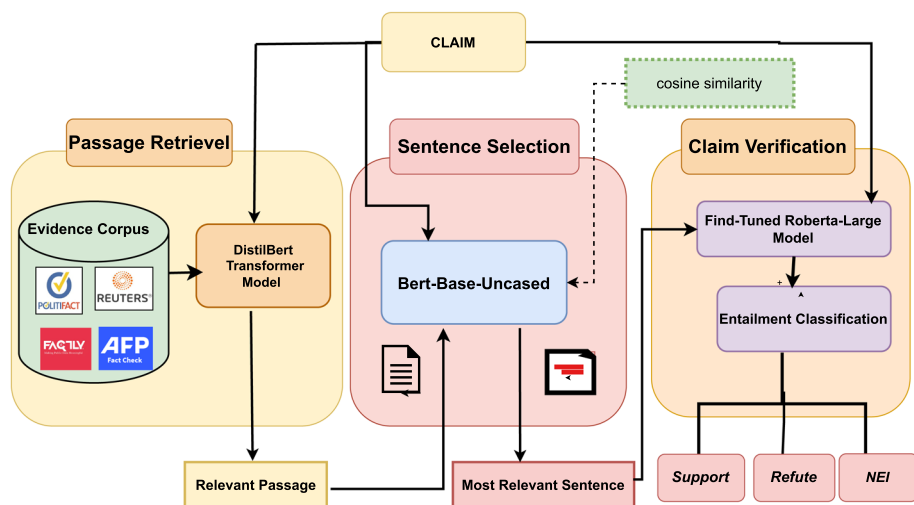


Fig. 4 Three components of the CREDIFY architecture illustrating the process of retrieving the most relevant passage and sentence for claim veracity prediction

4.1.1 Passage retrieval

The passage retrieval, the first module of our fact verification architecture, retrieves the most relevant passage from an evidence corpus. In contrast to previous approaches, which focused on retrieving evidence using term-frequency or keyword matching, CREDIFY implements context-aware and semantic-based passage retrieval evaluation with the assistance of powerful transformer models. The module takes two inputs, i.e., a claim and a whole evidence corpus, which consists of consistent expert-verified information. The transformer-based ms-macro-DistilBERT¹³ model converts the input claim and the passages in the corpus in a 768-dimensional dense vector space. Then, it calculates the degree of similarity between them individually. We utilize the cosine similarity function to determine the contextual and semantic proximity between a pair of claim and passage embedding. The claim and the evidence passage with a high similarity score is selected as the most relevant passage and is the input for the next stage. Equation 1 shows that the cosine distance between two input embedding, u and v , is the version of an inner product of vectors normalized by their L2 norms.

$$\text{CosSim}(u, v) = \frac{\sum_{i=1}^N u_i v_i}{\sqrt{\sum_{i=1}^N u_i^2} \sqrt{\sum_{i=1}^N v_i^2}} = \frac{\langle u, v \rangle}{\|u\| \|v\|} \quad (1)$$

where N indicates the number of dimensions with the sentence embedding representation u and v , $\langle u, v \rangle$ is an inner dot product between the two vectors. Our goal in this stage is to select the evidence available in the corpus that is highly related to the claim at the passage level and will be impactful for final veracity prediction. Figure 5 describes the passage retrieval process in detail.

¹³ <https://huggingface.co/sentence-transformers/msmarco-distilbert-base-tas-b>.

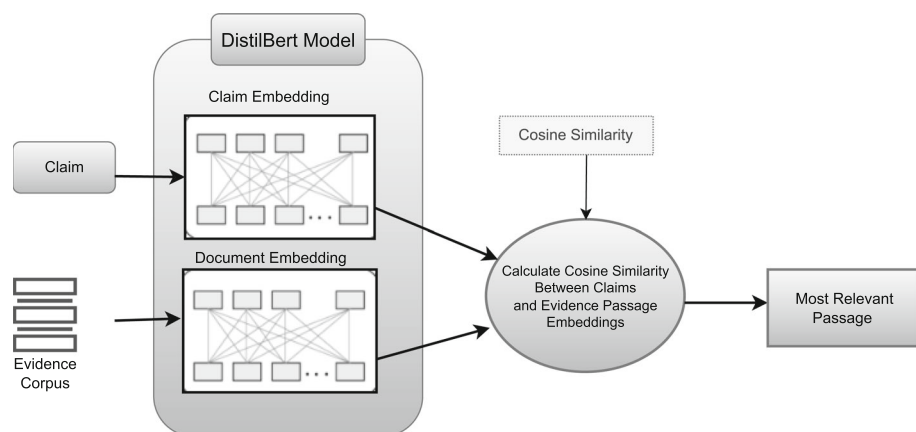


Fig. 5 The process of retrieving the most relevant passage

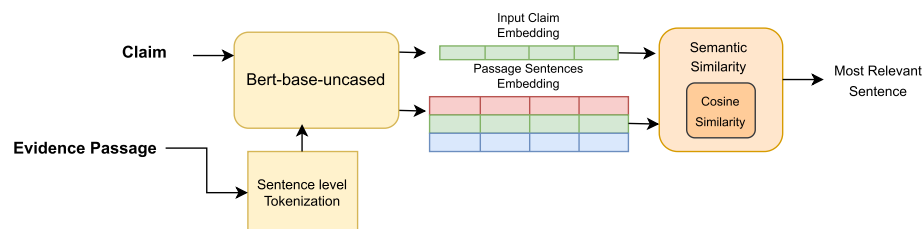


Fig. 6 Illustrating the process of retrieving the most relevant passage

4.1.2 Sentence selection

The second stage, sentence selection, uses the ‘bert-base-uncased’ model to extract the most relevant sentence from a passage retrieved in the previous stage. This relevant sentence acts as final evidence against the claim, which helps predict the veracity of the next step. The model splits the retrieved passage into sentences and then converts the passage and the claim into BERT embedding. The sentence with the highest similarity is selected as the most relevant sentence and powerful evidence for predicting the claim’s veracity. Figure 6 illustrates finding the most pertinent sentence.

4.1.3 Verdict prediction

Once the top relevant sentence is extracted, NLI techniques, or entailment classification, are applied to infer the relationship between the extracted evidence sentence (hypothesis) and the claim statement (premise). While semantic and contextual similarities cannot detect these nuances finer, the NLI model can detect a contradiction or entailment relationship between the sentences. We use the RoBERTa-base model that takes two inputs: an evidence sentence and a claim statement. The transformer-based models for NLI tasks often face challenges when using unseen domains, necessitating the adjustment of fine-tuning our model, enabling it to capture better the unique patterns and contexts present in the new and previously encountered data. This stage is the core component of our automated fact verification architecture. The model first converts the claim and evidence sentence into its embedding for capturing seman-

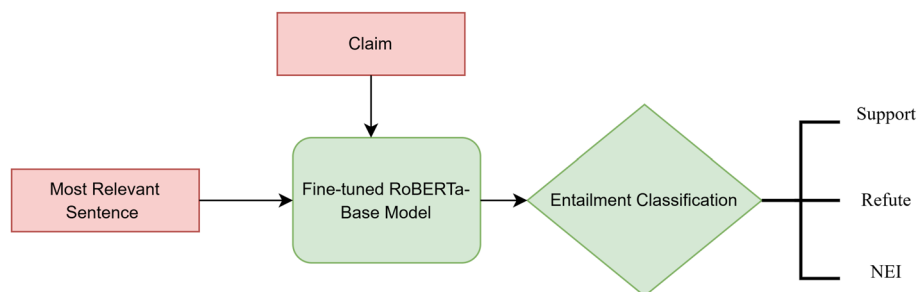


Fig. 7 RoBERTa-base model for fact verification task

tics. After this, the entailment classification technique analyzes the relationship between the claim and retrieved evidence, which then outputs the label $y \in \{\text{support, refute, NEI}\}$. Figure 7 illustrates the process of veracity prediction.

4.2 Data augmentation: contrastive claim generation and paraphrasing

This section presents the data augmentation techniques used for experimentation. First, we use a contrasting claim generation technique, which effectively increases the dataset scale and de-biases it by generating negative examples. These biases are typically present in datasets as there are patterns in a claim strongly associated with target labels and avoid performing proper factual reasoning over the evidence. For example, positive claims with words such as ‘failed to’ and ‘did not’ in the claim highly indicate the refute label because negative claims often have this pattern and may ignore the original evidence in the dataset. We use the CrossAug [55] data augmentation technique, which uses a fine-tuned BART model. Secondly, we use paraphrasing for data augmentation. We use a pre-trained T5 model trained on the ChatGPT paraphrase dataset Vladimir Vorobev [56]. The original claim is given to the T5 model, which generates the augmented claims. The next section presents the experimental setup and results.

5 Experimental setup and results

We evaluate the proposed methodology on our RECTIFY dataset at four levels of granularity: passage retrieval level, veracity prediction level, end-to-end framework, and data augmentation. For passage-level evaluation, we assess the performance of our transformer msmacro-DistilBERT model to retrieve the gold evidence and compare it with state-of-the-art baseline models. We do not fine-tune this model as it is specifically trained on information retrieval tasks like passage retrieval. The claims with labels NEI do not have any gold standard passage, so we ignored them in the passage level evaluation experiment. The veracity prediction is conducted to analyze the effect of transfer learning (fine-tuning or not) on NLI models. The complete end-to-end architecture is also evaluated, where we determine the performance of our framework CREDIFY, from retrieving relevant evidence to predicting final veracity with transformer models. Finally, the last experiment is performed with data augmentation contrastive claim generation technique to analyze the effect on contrastive learning. Furthermore, we also conduct our experiments using fivefold cross-validation and include

time performance to better understand the model's efficiency and scalability for real-world applications. In the following sections, we provide the experimental setup for implementing CREDIFY architecture and discuss the results of the models.

5.1 Experimental setup

This section discusses the details for implementing our proposed architecture CREDIFY. We use the Google Colab platform with Python 3.7 installed and T4 GPU for training our deep learning models. The full pipeline is implemented in three stages -*passage retrieval*, *sentence selection*, and *entailment classification*. For passage retrieval, we use TF-IDF, BM25, DPR-DistilRoBERTa, and the msMarco-DistilBERT model. The TF-IDF and BM25 hyperparameter values are set to default. Meanwhile, DPR-DistilRoBERTa and msMacro-DistilBERT are used with pre-trained settings and do not require fine-tuning. We use a performance metric called MRR (mean reciprocal rank) to check how well our passage retrieval model performs. MRR provides insights into the model's ability to accurately rank the correct passage and determine its position within the ranked list.

The second stage of sentence selection for extracting the most relevant sentence uses the 'bert-base-uncased'¹⁴ model. We do not train or fine-tune the model as it is already trained on BookCorpus, a dataset comprising 11,038 books and Wikipedia (excluding tables, lists, and headers). It is trained on 4 TPUs with 16 TPU chips for performing one million steps. The Adam optimizer with a learning rate of $1e - 4$ and a batch size (256) was set as a hyper-parameter for the training model.

The last stage, which predicts the claim's veracity, uses a pre-trained RoBERTa-base transformer model on five datasets: BookCorpus, English Wikipedia, CC-news, OpenWebText, and Stories. We further fine-tuned this model for our task because the pre-trained RoBERTa-base was trained on diverse texts. We fine-tune this model for NLI tasks with a sequence length of 512 and AdamW optimizer. The learning rate is set to $2e-5$, with a batch size of 16 and training for five epochs. To evaluate the performance of the models, we split our dataset and reserved 500 examples for testing, while the remaining 4500 examples were used for training.

To analyze the effect of transfer learning, we conducted an experiment that shows the effect of fine-tuning on the veracity prediction module. We give two inputs, claim and evidence sentences, to our pre-trained model and fine-tuned the model for our task, which predicts the veracity label by performing NLI techniques. We fine-tune the models by updating their weights with a sequence length of 512 and AdamW optimizer. We set the learning rate $2e-5$ with a batch size 16 and epoch 5.

The contrastive claim generation experiment is mainly performed with the CrossAug fine-tuned BART model at the veracity prediction stage. The hyper-parameter configuration for implementing this model includes setting the number of beams to 5 and the maximum length to 256. For paraphrasing the pre-trained T5 model, we use a maximum length of 128, beams of 5, and, a temperature of 0.9. Table 6 presents the hyperparameter values for experiments performed on CREDIFY.

To ensure a robust evaluation of our model, we conduct fivefold cross-validation. The dataset is split into 5 equal parts, where in each iteration four folds are used for training and one fold for testing. Each fold is rotating until all folds serve as the test set. This approach helps us avoid overfitting and provides a more generalized performance estimate. Additionally,

¹⁴ <https://huggingface.co/bert-base-uncased>.

Table 6 Hyperparameter values for proposed fact verification model

Module	Model	Hyperparameters
Passage retrieval	msmarco-DistilBERT TFIDF BM25 DPR	Learning rate of 7×10^{-6} max_df=1, min_df=1 b=0.75, k=1.2 learning rate of 6×10^{-4}
Sentence selection	bert-base-uncased	Learning rate: of 1×10^{-4} Batch Size:256 Optimizer: AdamW
Veracity prediction	RoBERTa-base BERT-base RoBERTa-Large RoBERTa-base-mnli Electra-base	Learning rate of 2×10^{-5} Sequence Length: 512 Optimizer: AdamW Batch size:16 Epoch: 5
Cross augmentation	fine-tuned BART	Num_beams = 5 Max_length = 256
Paraphrasing	pre-trained T5	Num_beams = 5 Max_length = 128 Temperature = 0.9

we also conduct a time performance analysis on all five folds to assess the computational efficiency of our model across different train and test splits.

5.2 Evaluation metrics

Similar to previous work on fact verification [15, 34], we use accuracy, micro precision, recall, F1 score, and mean reciprocal rank (MRR) to evaluate the performance of our proposed methodology. The mathematical expressions for each evaluation metric are presented in Eq. 2.

$$\text{Metrics} = \begin{cases} \text{Accuracy} = \frac{T_P + T_N}{T_P + T_N + F_P + F_N} \\ \text{Micro-Precision} = \frac{\sum_i T_{P_i}}{\sum_i T_{P_i} + \sum_i F_{P_i}} \\ \text{Micro-Recall} = \frac{\sum_i T_{P_i}}{\sum_i T_{P_i} + \sum_i F_{N_i}} \\ \text{Micro-F1} = 2 \cdot \frac{\text{Micro-Precision} \cdot \text{Micro-Recall}}{\text{Micro-Precision} + \text{Micro-Recall}} \\ \text{MRR@k} = \frac{1}{\text{rank}}, \text{ if correct passage is in top } k, \text{ else } 0. \end{cases} \quad (2)$$

5.3 Baselines

To evaluate CREDIFY architecture's performance, we compare with the following baselines established by Samarinas et al. [34]; Khan et al. [30]; DeHaven [15].

- Quin+ [34] uses BM25 for retrieving evidence passage from top web results and Bert sentence selection approach along with RoBERTa-Large as NLI technique for verifying open-domain claims from Factual-NLI+ dataset
- WatClaimCheck [30] uses the DPR model for retrieving evidence passage from justification articles and RoBERTa-base as an NLI approach for verifying open-domain claims.
- BEVERS [15] uses TF-IDF for retrieving evidence and RoBERTa-Large as an NLI approach. They implement their methodology on the FEVER dataset to verify Wikipedia claims.

Table 7 Evaluation of passage retrieval module, reporting the MRR percentage

Algorithms	Retrieval mechanism	MRR@3
BM25	Relevance score	87%
TF-IDF	Term frequency	89%
DPR-DistilRoBERTa	Semantic and context	91%
msmarco-DistilBERT	Semantic and context	94%

6 Results

This section reports the results of our experiments, highlighting the performance of the CREDIFY architecture across various performance metrics.

6.1 Passage retrieval evaluation

We compare the performance of the ‘msmarco-DistilBERT’ model, which has been effectively used for question-answering systems before, with the baseline models DPR-DistilRoBERTa base [30], TF-IDF [15] and BM25[19]. Table 7 demonstrates that the utilization of msmarco-DistilBERT for fact verification evidence retrieval task surpasses the performance of the baseline models with an MRR@3 of 94%. However, BM25 and TF-IDF perform worst among all the models due to a limited understanding of semantic and contextual relationships within the text.

6.2 Veracity prediction module evaluation

In this experiment, we analyze the impact of transfer learning by fine-tuning our models on the veracity prediction module. We experimented with two techniques. First, we evaluate the role of the pre-trained language models by freezing the layers during the training phase. Secondly, we assess the impact of fine-tuning the model by adjusting and updating its weights based on the new task-specific data. The entire pre-trained model is trained on gold sentence evidence and the claim statement. The output of the model is fed to neural network layers, followed by a softmax layer. The error is back-propagated, and the pre-trained weights of the models are updated based on the newly proposed dataset. Table 8 demonstrates the comparative results of these two techniques. The experimental results clearly show that fine-tuning significantly impacts the performance of the veracity prediction module. For instance, models like bert-base-uncase perform worst among all the models, but a significant increase in all the performance metrics is seen by fine-tuning it. We also analyze the impact of DeBERTav3-mnli-snli-anli and RoBERTa-Large-mnli models, which are trained on popular datasets NLI, SNLI, and ANLI. These models also cannot achieve good performance without fine-tuning due to the high bias in these datasets. Further, these datasets consist of concise one-line sentences. Consequently, models trained on these three datasets usually do not perform efficiently on real-world tasks. However, among all the models (fine-tuned and without fine-tuned), the fine-tuned RoBERTa-base model performs best with an accuracy of 97%. These results suggest that fine-tuning can enhance the performance of certain models.

Table 8 The results of the NLI models with and w/o fine-tuning

Models	Metrics			
	Precision	Recall	F1 score	Accuracy
bert-base-uncased ^(a)	0.38	0.28	0.38	0.23
bert-base-uncased	0.85	0.83	0.84	0.84
DeBERTav3-mnli-snli-anli ^(a)	0.31	0.41	0.33	0.41
DeBERTav3-mnli-snli-anli	0.51	0.54	0.61	0.65
DeBERTa-large ^(a)	0.78	0.75	0.79	0.91
DeBERTa-large	0.93	0.89	0.95	0.96
RoBERTa-large-mnli ^(a)	0.48	0.61	0.53	0.41
RoBERTa-large-mnli	0.54	0.45	0.39	0.41
RoBERTa-base ^(a)	0.89	0.84	0.85	0.92
RoBERTa-base	0.97	0.94	0.95	0.97

^(a) Indicates models with frozen weights

6.2.1 End-to-end architecture evaluation

We evaluated and assessed our proposed end-to-end architecture comprising three modules and compared its performance with previously proposed fact-checking models. The collective functionality of these modules involves retrieving the most relevant passage and, subsequently, the relevant sentence from it. This extracted sentence plays a pivotal role in predicting the veracity of a given claim. Finally, the in-charge veracity prediction module analyzes the relationship between the claim and the retrieved sentence. This analysis classifies the relationship as support, refute, or NEI. The model details and their hyperparameter tuning are discussed in Table 6. We perform experiments on our proposed dataset RECTIFY, which is collected explicitly for multidomain automated claim verification.

The results of CREDIFY experiments are discussed in Table 9, showcasing the performance evaluation of six state-of-the-art pre-trained veracity prediction models with msmacro-DistilBERT and bert-base-uncase as evidence (passage and sentence) retrieval models. The pretrained veracity prediction models include bart-large-mnli, GPT-3.5, bert-base-uncase, RoBERTa-Large, DeBERTa-base-mnli, RoBERTa-base. Notably, the RoBERTa-base model performs best with an accuracy of 96.5%, micro-precision of 98.12%, micro-F1 score of 95.45%, and micro-recall of 93.25%. However, large language models GPT-3 and bart-large-mnli perform worst among all the models due to a lack of true understanding and evidence-based reasoning abilities. As discussed above, the pre-trained models on two popular NLI datasets (Multi-NLI and SNLI) are not effective in a real-world scenario due to present biases and generalization ability. Therefore, the pre-trained model DeBERTa-base-mnli also performs poorly on our task.

6.2.2 Data augmentation evaluation and results

The effect of data augmentation contrastive claim generation technique for de-biasing and generalizing our fact verification models is also analyzed. We have increased the scale of negative claims by contrastive conversion of positive claims in our model as discussed in Sect. 4.2. This contrastive augmentation technique increases the scale, capturing better insights and gen-

Table 9 Evaluation results on the end-to-end framework for automated fact verification with DistilBERT and BERT as evidence retrieval technique with fine-tuned NLI models

Model	Micro-precision(%)	Micro-recall(%)	Micro-F1 score(%)	Accuracy(%)
bart-large-mnli	10.12	32.56	16.93	32.43
GPT-3.5	79.14	78.24	78.85	80.49
BERT base model (uncased)	92.12	93.58	91.80	89.35
RoBERTa-Large	93.61	93.94	93.27	93.57
DeBERTa-base	93.23	94.76	98.01	92.87
RoBERTa-base	98.12	93.55	95.45	96.55

Table 10 Results with and w/o augmentation with RoBERTa-base model on support, refute, and NEI classes

Model	Precision(%)	Recall(%)	F1 score(%)
<i>With cross-augmentation</i>			
Support	0.93	0.80	0.86
Refute	0.92	0.97	0.90
NEI	0.88	0.92	0.90
<i>With paraphrasing</i>			
Support	0.94	0.99	0.97
Refute	0.83	0.99	0.91
NEI	0.99	0.94	0.97
<i>Without augmentation</i>			
Support	0.94	0.98	0.96
Refute	0.89	0.92	0.90
NEI	0.97	0.93	0.95

Bold values indicate the highest F1 score for each class

eralizing the patterns of negative claims. Furthermore, we also apply to paraphrase for data augmentation using the T5¹⁵ model. In this experiment, we examine how contrastive claim generation and paraphrasing impact fact verification veracity label prediction, comparing it to the standard method without a data augmentation model. We only use RoBERTa-base model for performing this experiment. The results indicate that paraphrasing performs best for all the labels with 97 % F1 score for refute and 99% F1 score for support and NEI classes. Table 10 reports the results with and without augmentation using cross-augmentation and paraphrasing.

6.2.3 Cross-validation and time analysis

Our fivefold cross-validation results demonstrate consistent performance across all folds, with a validation accuracy ranging from 91.67% to 95.17% while the evaluation accuracy varies between 92.90% and 95.60% with an average accuracy of 93.84%. This shows the scalability and reliability of our veracity prediction model. The evaluation time remains stable across all folds averaging around 1319 s per fold, suggesting uniform computational efficiency throughout the process. These results suggest that the model maintains reliable predictive performance while consistent training time per fold, potentially due to computational variations. Table 11 presents the cross-validation results and time performance on each fold.

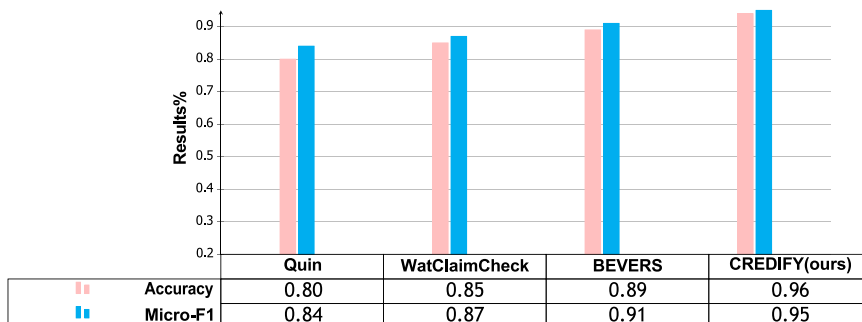
6.3 Discussion

Our proposed methodology for verifying claims on the RECTIFY dataset surpasses the performance of Khan et al. [30] by a margin of 5.5% and achieves an accuracy score of 96.5%. Figure 8 compares the performance of our proposed methodology with previous research conducted in fact verification. The comparative analysis reveals our approach's superiority and

¹⁵ https://huggingface.co/humarin/chatgpt_paraphraser_on_T5_base.

Table 11 Results of fivefold cross-validation and time analysis

Folds	Validation accuracy	Evaluation accuracy	Evaluation time (s)
1	0.9433	0.9330	1316.19
2	0.9458	0.9290	1320.22
3	0.9167	0.9310	1320.26
4	0.9458	0.9430	1319.77
5	0.9517	0.9560	1319.73
Average accuracy	0.9384 ± 0.0100		

**Fig. 8** The comparison of our proposed methodology with previous studies on the RECTIFY dataset [15, 30, 34]

valuable insights contributing to achieving these results. Our proposed framework improves the automated fact verification process for the following reasons:

- We aim to train a model that efficiently handles open-domain claim areas by focusing on gathering claims from various subjects rather than targeting one specific claim topic.
- We collect the evidence consistently and discourage the real-time retrieval of evidence from top results, which helps avoid factual inconsistency problems.
- We rely on context and semantics-based evidence passage retrieval technique with the msmacro-DistilBERT model, which is specifically designed for passage retrieval tasks with no training data needed.
- We use already pre-trained models for veracity prediction and fine-tune them on our task, encouraging transfer learning instead of relying on initial model weights.
- The baseline dataset [30] consolidates 86 labels into three broad categories without providing clear annotation guidelines, which may introduce ambiguity and subjectivity in label assignment. In contrast, the proposed RECTIFY dataset carefully addresses this issue by ensuring that veracity labels are assigned exclusively by experts, thereby enhancing reliability and reducing subjectivity.

In Table 12, we highlight the approaches used by previous studies Khan et al. [30]; DeHaven [15] and compare them with our proposed framework. The evidence retrieval technique they use either retrieves evidence based on term frequency or requires large training data for efficient retrieval. However, in our methodology, the pre-trained msmacro-DistilBERT model retrieves evidence by considering context, and it does not require fine-tuning, so it performs well on small data. This efficient evidence passage retrieval affects the overall veracity

Table 12 Comparison with previous studies on automated fact verification system

References	Evidence retrieval	Entailment classification	Evidence type	Accuracy
Samarinas et al. [34]	BM25	RoBERTa Large	Articles	80.1%
Khan et al. [30]	DPR	RoBERTa base	Article	85.6%
DeHaven [15]	TF-IDF	RoBERTa Large	Wikipedia documents	89.9%
Proposed methodology	mmacro-Distil- BERT	RoBERTa- base	Articles	96.5%

prediction using state-of-the-art techniques and performs best compared to baseline with an accuracy of 96.5%.

While CREDIFY presents a significant approach to automated claim verification, the framework has several limitations. Our study found that msmacro-DistilBERT outperforms other baseline models for evidence retrieval, which is indeed an exciting finding. However, future directions could explore whether these results are primarily from the model being trained on a macro dataset or if factors such as hyperparameter tuning of the models contribute to this significant impact. Furthermore, while the contrastive claim generation experiment undoubtedly proved beneficial in reducing bias and enhancing the performance of negative claim veracity prediction, it is essential to acknowledge its potential limitation. Specifically, increasing the scale and prioritizing negative claims also leads to data imbalance problems. This imbalance could introduce challenges in model training and evaluation, potentially compromising the effectiveness and fairness of the claim veracity prediction process. Finally, as the information is generated in many shapes and forms, including text, video, and audio, due to the rapid increase in big data, our proposed architecture CREDIFY is limited in handling the veracity prediction of only textual data claims.

7 Conclusion

This paper makes three key contributions to the field of evidence retrieval and fact verification. First, it introduces RECTIFY, a dataset designed for automated fact verification. This dataset consists of open-domain claims paired with expert-written justification articles sourced from leading fact-checking websites. The justification articles serve as evidence, written by professional fact-checkers to support their claims. Second, this study introduces an end-to-end framework, CREDIFY, comprising three stages: passage retrieval, sentence selection, and veracity prediction. The pre-trained msmarco-DistilBERT model is utilized in the passage retrieval stage to ensure semantic and context-based evidence retrieval, achieving the highest MRR of 94%. A pre-trained BERT model was employed to identify the most relevant sentence, which served as the final piece of evidence. Lastly, the RoBERTa-base model was fine-tuned to assess the relationship between the claim and the extracted evidence sentence, ultimately predicting the claim's veracity. The third key contribution of this study is data augmentation through paraphrasing, which improved performance across all classes. The results show that the fine-tuned RoBERTa-base model achieved the highest accuracy of 97% when evaluated on gold-standard evidence. The proposed end-to-end framework, CREDIFY, achieved an accuracy of 96.5%, outperforming state-of-the-art methods by a significant margin of 5.5%. To address the class imbalance issue caused by contrastive claim generation, we employed a paraphrase-based data augmentation approach, which balanced all classes and enhanced overall performance.

8 Limitations and future work

Our proposed model shows promising results in passage retrieval and claim verification using transformer-based models, such as msmarco-DistilBERT and RoBERTa-base. However, certain limitations need to be addressed. Our dataset comprises 5,000 claims extracted from various news and fact-checking websites, ensuring the reliability of claim verification.

However, the dataset does not provide additional information from social networks, where misinformation tends to be more informal and context-dependent.

While transformer-based models improve retrieval accuracy, false positives and negatives remain, particularly for claims requiring external contextual knowledge. We employ two data augmentation techniques to handle the data imbalance issue: contrastive claim generation and paraphrasing. Although these methods enhance the model's robustness, contrastive claim generation is crucial and may introduce overfitting. Despite the effectiveness of transformer-based models, they remain a black-box system, making it difficult to interpret why a claim is classified as support, refute, or NEI. Transformer-based models, particularly Roberta, with millions of parameters, require significant computational resources for fine-tuning and inference. Although the dataset size is relatively smaller; we could not train transformer-based models on larger data sets due to the lack of computational resources.

In the future, we plan to expand the dataset to include social media sources such as Facebook and X, enhancing the model's robustness. Additionally, we will explore advanced data augmentation techniques to improve data quality and transformation. Furthermore, we will investigate the key factors contributing to the superior performance of the msmarco-DistilBERT model compared to other baseline models. In addition, social context and user profiling will be integrated to improve the performance of fact verification. Furthermore, we will adapt network immunization and explore adaptive algorithms such as CONTAIN, SparseShield, and MCWDST for real-time fake news detection. Another future direction involves enhancing document similarity by incorporating clustering techniques for more effective claim retrieval.

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Author Contributions All authors contributed equally to the writing of this manuscript.

Data availability The research data supporting the results of this manuscript can be accessed at the following GitHub repository: <https://github.com/SarahNasir54/Credify/tree/main/Credify>.

Declarations

Conflict of interest The authors declare no conflict of interest.

References

1. Rodríguez-Ferrándiz R (2023) An overview of the fake news phenomenon: from untruth-driven to post-truth-driven approaches. *Media Commun* 11(2):15–29
2. Hassan N, Arslan F, Li C, Tremayne M (2017) “Toward automated fact-checking: Detecting check-worthy factual claims by claimbuster.” In: *Proceedings of the 23rd ACM SIGKDD international conference on knowledge discovery and data mining*, pp. 1803–1812
3. Alhindi T, Petridis S, Muresan S (2018) “Where is your evidence: Improving fact-checking by justification modeling.” In: *Proceedings of the first workshop on fact extraction and verification (FEVER)*, pp. 85–90
4. Allen J, Arechar AA, Pennycook G, Rand DG (2021) Scaling up fact-checking using the wisdom of crowds. *Sci Adv* 7(36):eabf4393
5. Hassan N, Zhang (2017) Claimbuster: the first-ever end-to-end fact-checking system. *Proc VLDB Endowment* 10(12):1945–1948
6. Hughes HC, Waismel-Manor I (2021) The macedonian fake news industry and the 2016 us election. *PS: Political Sci Amp Politics* 54(1):19–23
7. Dhiman P, Kaur A, Iwendi C, Mohan SK (2023) A scientometric analysis of deep learning approaches for detecting fake news. *Electronics* 12(4):948

8. Budak C (2016) What happened? the spread of fake news publisher content during the, us presidential election In: *The World Wide Web Conference 2019*:139–150
9. Grossman GM, Helpman E (2023) Electoral competition with fake news. *Eur J Polit Econ* 77:102315
10. Martín A, Huertas-Tato J, Huertas-García Álvaro, Villar-Rodríguez G, Camacho D (2022) “Facter-check: Semi-automated fact-checking through semantic similarity and natural language inference,” *Knowledge-Based Systems*, vol. 251, p. 109265. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0950705122006323>
11. Thorne J, Vlachos A (2018) “Automated fact checking: Task formulations, methods and future directions,” In: *Proceedings of the 27th International Conference on Computational Linguistics*, pp. 3346–3359
12. Guo Z, Schlichtkrull M, Vlachos A (2022) A survey on automated fact-checking. *Trans Assoc Comput Linguistics* 10:178–206
13. Atanasova P, Simonsen JG, Lioma C, Augenstein I (2020) “Generating fact checking explanations,” In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 7352–7364
14. Hanselowski A, Stab C, Schulz C, Li Z, Gurevych I (2019) “A richly annotated corpus for different tasks in automated fact-checking,” In: *Proceedings of the 23rd Conference on Computational Natural Language Learning (CoNLL)*, pp. 493–503
15. DeHaven (2023) “Bever: A general, simple, and performant framework for automatic fact verification,” in *Proceedings of the Sixth Fact Extraction and VERification Workshop (FEVER)*, pp. 58–65
16. Evans M, Soós D, Landers E, Wu J (2023) “Msvec: A multidomain testing dataset for scientific claim verification,” in *Proceedings of the Twenty-fourth International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing*, pp. 504–509
17. Thorne J, Vlachos A, Christodoulopoulos C, Mittal A (2018) “Fever: a large-scale dataset for fact extraction and verification,” In: *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, pp. 809–819
18. Liu Z, Xiong C, Dai Z, Sun S, Sun M, Liu Z (2020) Adapting open domain fact extraction and verification to covid-fact through in-domain language modeling. *Findings Assoc Comput Linguistics: EMNLP 2020*:2395–2400
19. Bouziane M, Perrin H, Sadeq A, Nguyen T, Cluzeau A, Mardas J (2021) “Fabulous: Fact-checking based on understanding of language over unstructured and structured information,” In: *Proceedings of the Fourth Workshop on Fact Extraction and VERification (FEVER)*, pp. 31–39
20. Gupta P, Wu C-S, Liu W, Xiong C (2022) “Dialfact: A benchmark for fact-checking in dialogue,” In: *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 3785–3801
21. Hanselowski A, Stab C, Schulz C, Li Z, Gurevych I (2019) “A richly annotated corpus for different tasks in automated fact-checking,” In: *Proceedings of the 23rd Conference on Computational Natural Language Learning (CoNLL)*, pp. 493–503
22. Hassan, Naemul, Zhang (2017) “Claimbuster: The first-ever end-to-end fact-checking system,” *Proceedings of the VLDB Endowment*, vol. 10, no. 12, pp. 1945–1948
23. Sathe A, Ather S, Le TM, Perry N, Park J (May 2020) “Automated fact-checking of claims from Wikipedia,” In: *Proceedings of the Twelfth Language Resources and Evaluation Conference*. Marseille, France: European Language Resources Association, pp. 6874–6882. [Online]. Available: <https://aclanthology.org/2020.lrec-1.849>
24. Diggelmann T, Boyd-Graber J, Bulian J, Ciaramita M, Leippold M (2020) “Climate-fever: A dataset for verification of real-world climate claims,” *arXiv preprint arXiv:2012.00614*
25. Alhindi T, Petridis S, Muresan S (2018) “Where is your evidence: Improving fact-checking by justification modeling,” In: *Proceedings of the first workshop on fact extraction and verification (FEVER)*, pp. 85–90
26. Kotonya N, Toni F (2020) “Explainable automated fact-checking for public health claims,” In: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 7740–7754
27. Wadden D, Lin S, Lo K, Wang LL, van Zuylen M, Cohan A, Hajishirzi H (2020) “Fact or fiction: Verifying scientific claims,” In: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 7534–7550
28. Rangapur A, Wang H, Shu K (2023) “Fin-fact: A benchmark dataset for multimodal financial fact checking and explanation generation,” *arXiv preprint arXiv:2309.08793*
29. Nadeem M, Fang W, Xu B, Mohtarami M, Glass J (2019) “Fakta: An automatic end-to-end fact checking system,” In: *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (Demonstrations)*, pp. 78–83
30. Khan K, Wang R, Poupart P (2022) “Watclaimcheck: A new dataset for claim entailment and inference,” In: *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 1293–1304

31. Fan A, Piktus A, Petroni F, Wenzek G, Saeidi M, Vlachos A, Bordes A, Riedel S (2020) "Generating fact checking briefs," In: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), pp. 7147–7161
32. Ennen P, Freddi F, Lin C-J, Kung P-N, Wang C-Y, Yang D-s, Shiu, Bernacchia A (2023) "Hierarchical representations in dense passage retrieval for question-answering," In: Proceedings of the Sixth Fact Extraction and VERification Workshop (FEVER). Dubrovnik, Croatia: Association for Computational Linguistics, May, pp. 17–28. [Online]. Available: <https://aclanthology.org/2023.fever-1.2>
33. Sathe A, Park J (2021) "Automatic fact-checking with document-level annotations using bert and multiple instance learning," In: Proceedings of the Fourth Workshop on Fact Extraction and VERification (FEVER), pp. 101–107
34. Samarinas C, Hsu W, Lee ML (2021) "Improving evidence retrieval for automated explainable fact-checking," In: Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies: Demonstrations, pp. 84–91
35. Reddy AJ, Rocha G, Esteves D (2018) "Defactonlp: Fact verification using entity recognition, tfidf vector comparison and decomposable attention," In: Proceedings of the First Workshop on Fact Extraction and VERification (FEVER), pp. 132–137
36. Truică C-O, Apostol E-S, Paschke A (2022) Awakened at checkthat!, Fake news detection using bilstm and sentence transformer. CEUR Workshop Proc 3180(2022):749–757
37. Pradeep R, Ma X, Nogueira R, Lin J (2021) "Scientific claim verification with vertSerini," In: Proceedings of the 12th International Workshop on Health Text Mining and Information Analysis, pp. 94–103
38. Truică C-O, Apostol E-S, Karras P (2024) Danes: Deep neural network ensemble architecture for social and textual context-aware fake news detection. Knowl-Based Syst 294:111715
39. Radu R-G, Rădulescu I-M, Truică C-O, Apostol E-S, Mocanu M (2020) Clustering documents using the document to vector model for dimensionality reduction, In: ieeee international conference on automation, quality and testing, robotics (aqtr). IEEE 2020:1–6
40. Rădulescu I-M, Boicea A, Truică C-O, Apostol E-S, Mocanu M, Rădulescu F (2021) "Denlac: Density levels aggregation clustering—a flexible clustering method—," In: International Conference on Computational Science. Springer, pp. 316–329
41. Apostol E-S, Truică C-O, Paschke A (2024) "Contcommrtd: A distributed content-based misinformation-aware community detection system for real-time disaster reporting," IEEE Transactions on Knowledge and Data Engineering
42. Truica C-O, Leordeanu CA (2017) Classification of an imbalanced data set using decision tree algorithms. Univ Politech Bucharest Sci Bull Ser C Electr Eng Comput Sci 79:69–84
43. Truica C-O, Apostol ES, Leordeanu CA (2017) "Topic modeling using contextual cues," In: 19th International Symposium on Symbolic and Numeric Algorithms for Scientific Computing (SYNASC). IEEE 2017:203–210
44. Sadeghi F, Bidgoly AJ, Amirkhani H (2022) Fake news detection on social media using a natural language inference approach. Multimedia Tools Appl 81(23):33 801–33–821
45. Pan L, Chen W, Xiong W, Kan M-Y, Wang WY (2021) "Zero-shot fact verification by claim generation," In: Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 2: Short Papers), pp. 476–483
46. Ilie V-I, Truică C-O, Apostol E-S, Paschke A (2021) "Context-aware misinformation detection: A benchmark of deep learning architectures using word embeddings," IEEE Access, vol. 9, pp. 162 122–162 146
47. Truică C-O, Apostol E-S (2022) Misrobarta: transformers versus misinformation. Mathematics 10(4):569
48. Truică C-O, Apostol E-S (2023) It's all in the embedding! fake news detection using document embeddings. Mathematics 11(3):508
49. Petrescu A, Truică C-O, Apostol E-S, Karras P (2021) "Sparse shield: Social network immunization vs. harmful speech," In: Proceedings of the 30th ACM International Conference on Information & Knowledge Management, pp. 1426–1436
50. Truică C-O, Apostol E-S, Nicolescu R-C, Karras P (2023) "Mcwdst: a minimum-cost weighted directed spanning tree algorithm for real-time fake news mitigation in social media," IEEE Access, vol. 11, pp. 125 861–125 873
51. Apostol ES, Coban Ö, Truică C-O (2024) Contain: a community-based algorithm for network immunization. Eng Sci Technol Int J 55:101728
52. Rădulescu I-M, Truică C-O, Apostol E-S, Dobre C (2020) "Enhancing scientific collaborations using community detection and document clustering," In: 2020 IEEE 16th International Conference on Intelligent Computer Communication and Processing (ICCP). IEEE, pp. 43–50
53. Augenstein I, Lioma C, Wang D, Lima LC, Hansen C, Hansen C, Simonsen JG (2019) "Multifc: A real-world multi-domain dataset for evidence-based fact checking of claims," In: Proceedings of the

- 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), pp. 4685–4697
54. Chen S, Khashabi D, Yin W, Callison-Burch C, Roth D (2019) “Seeing Things from a Different Angle: Discovering Diverse Perspectives about Claims,” In: Proc. of the Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL). [Online]. Available: <http://cogcomp.org/papers/CKYCR19.pdf>
 55. Lee M, Won S, Kim J, Lee H, Park C, Jung K (2021) “Crossaug: A contrastive data augmentation method for debiasing fact verification models,” In: Proceedings of the 30th ACM International Conference on Information & Knowledge Management, pp. 3181–3185
 56. Vorobev MK Vladimir (2023) “Chatgpt paraphrases dataset,”

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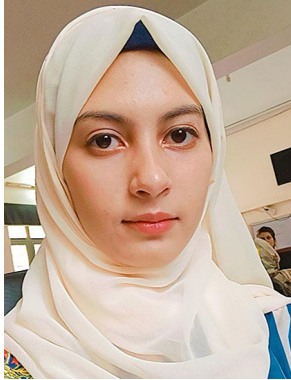
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